

Zero-Knowledge Proofs

Lecture 22 — Sigma Protocols, Fiat–Shamir, SNARKs and STARKs

SEC 595 — Encrypted Computing

- 1 Proof Systems
- 2 Sigma Protocols and Schnorr
- 3 Commitments
- 4 Fiat–Shamir
- 5 Arithmetisation
- 6 Polynomial Commitments
- 7 SNARKs and STARKs

Relation

A relation $\mathcal{R}$ is a set of pairs (x, w) : x the **statement**, w the **witness**, membership checkable in polynomial time. $L_{\mathcal{R}} = \{x : \exists w, (x, w) \in \mathcal{R}\}$.

- The **prover** P holds x and w .
- The **verifier** V holds only x , runs in polynomial time, and accepts or rejects.

Examples: $\mathcal{R}_{\text{DL}}, ((\mathbb{G}, g, h), x)$ with $h = g^x$; $\mathcal{R}_{\text{exec}}, ((\text{program}, \text{in}, \text{out}), \text{trace})$.

The three properties

Interactive proof system, soundness error ε

- ① **Completeness.** $(x, w) \in \mathcal{R} \Rightarrow V$ accepts. *protects the prover*
- ② **Soundness.** $x \notin L_{\mathcal{R}} \Rightarrow$ every P^* is accepted with probability $\leq \varepsilon$.
protects the verifier
- ③ **Zero knowledge.** A simulator $\mathcal{S}$ reproduces the verifier's view from x alone. *protects the prover*

Soundness against **all** provers: a **proof**. Against **efficient** provers only: an **argument**.

The “AR” in SN**ARK** and ST**ARK** is *argument*.

Zero knowledge is the simulation paradigm of Lecture 21:

for every verifier V^ there exists a simulator $\mathcal{S}$*

whose output, from x alone, is indistinguishable from the real view.

Whatever V^* computes from the transcript it could compute without the prover. So the transcript carried nothing else.

Grade	Real and simulated views are
perfect	identically distributed
statistical	at negligible statistical distance
computational	indistinguishable to efficient algorithms

In a cyclic group every h is a power of g : the language of $\mathcal{R}_{\text{DL}}$ is everything, and soundness is vacuous.

What we want is that the prover **knows** x .

Definition

(P, V) is a proof of knowledge if an efficient **extractor** $\mathcal{E}$, with *rewinding* access to any P^* that convinces V often enough, outputs a witness.

Rewinding: rerun P^* from a saved state with a different challenge.

The simulator produces a transcript *without* the witness; the extractor produces the witness *from* transcripts.

HVZK

$\mathcal{S}$ need only reproduce the view of a verifier that follows the protocol: its challenge is uniform and independent of the prover's first message.

Full ZK

$\mathcal{S}$ must succeed against *every* efficient V^* , including one that picks its challenge as a function of what it saw.

Fiat–Shamir makes the challenge a hash, so it *is* uniform and independent. That is why HVZK suffices in practice.

$P \longrightarrow V$: **commitment** a

$V \longrightarrow P$: **challenge** $c \leftarrow \mathcal{C}$, uniform

$P \longrightarrow V$: **response** z

Sigma protocol

- ① **Completeness.**
- ② **Special soundness:** from accepting (a, c, z) and (a, c', z') with the same a and $c \neq c'$, a witness can be computed efficiently.
- ③ **HVZK.**

Public coin: the verifier sends only randomness.

$\mathbb{G} = \langle g \rangle$ of prime order q ; statement $h = g^x$; witness $x \in \mathbb{Z}_q$.

Protocol

- 1 P samples $k \leftarrow \mathbb{Z}_q$, sends $a = g^k$.
- 2 V samples $c \leftarrow \mathbb{Z}_q$, sends it.
- 3 P sends $z = k + cx \bmod q$.
- 4 V accepts iff $g^z = a \cdot h^c$.

Completeness: $g^z = g^k (g^x)^c = a h^c$. Exponents live mod $q = \text{ord}(g)$.

k is a one-time pad: $z = k + cx$ is uniform and alone reveals nothing about x .

Theorem

From accepting (a, c, z) and (a, c', z') with $c \neq c'$,

$$x = (z - z')(c - c')^{-1} \bmod q.$$

Dividing $g^z = ah^c$ by $g^{z'} = ah^{c'}$ gives $g^{z-z'} = h^{c-c'}$; $c - c'$ is invertible since q is prime.

So a prover without x answers at most one challenge per a : soundness error $1/|\mathcal{C}|$.

$|\mathcal{C}| = 2^{128}$: one round suffices.

Theorem (perfect HVZK)

Given h and c , sample $z \leftarrow \mathbb{Z}_q$ and set

$$a = g^z h^{-c}.$$

Real: z uniform, a fixed by the verification equation. Simulated: the same. The distributions are **identical**.

The simulator chose z *first* and back-solved for a . A real prover cannot: it sends a before it sees c .

The order of moves is what separates a proof from a forgery.

Example — a Schnorr transcript

$\mathbb{G} = \langle 2 \rangle \subseteq \mathbb{F}_{23}^\times$, witness $x = 6$.

- a. Confirm $\text{ord}(2) = 11$. Which modulus do exponents live in?
- b. Compute the statement $h = g^x$.
- c. Run the protocol with $k = 4$ and $c = 7$: give a , z , and both sides of $g^z = ah^c$.

Same instance.

- a. **Rewind.** Same a , new challenge $c' = 2$. Compute z' , check it accepts, and run the extractor. Do you get $x = 6$?
- b. **Simulate.** With no witness: fix $c = 3$, take $z = 10$, and produce an accepting transcript.
- c. What is the soundness error of one round in this group?

$\text{Com}(m; r) \rightarrow c$; opening reveals (m, r) .

- **Hiding**: c reveals nothing about m .
- **Binding**: the committer cannot open c two ways.

Proposition

No scheme is both perfectly hiding and perfectly binding.

If c has the same distribution for every m , then some r' opens it to any m' , and an unbounded committer finds it.

Scheme	Hiding	Binding
Pedersen $g^m h^r$ Hash $H(m r)$	perfect computational	computational (DLog) collision resistance

Definition

$\mathbb{G}$ of prime order q , generators g, h with $\log_g h$ **unknown to the committer**, $m, r \in \mathbb{Z}_q$:

$$\text{Com}(m; r) = g^m h^r.$$

Perfectly hiding: for fixed m , h^r is uniform on $\mathbb{G}$, so $g^m h^r$ is uniform for every m .

Binding is DLog: $g^m h^r = g^{m'} h^{r'}$ with $m \neq m'$ gives

$$\log_g h = (m - m')(r' - r)^{-1} \bmod q.$$

Proposition

$$\text{Com}(m_1; r_1) \cdot \text{Com}(m_2; r_2) = \text{Com}(m_1 + m_2; r_1 + r_2),$$

both sums mod q .

A prover can prove **linear relations** among committed values without opening them.

To show $m_3 = m_1 + m_2$: the commitments differ by a commitment to zero, h^r — a Schnorr proof of knowledge of r .

Schnorr's $a = g^k$ plays the same role: it freezes k before the challenge.

$$\mathbb{G} = \langle 2 \rangle \subseteq \mathbb{F}_{23}^\times, \quad g = 2, \quad h = g^7 = 13.$$

- a. Compute $\text{Com}(5; 3)$ and $\text{Com}(4; 9)$. Which modulus at each step?
- b. Multiply them. Which opening does the product have? Check it.
- c. Check that $\text{Com}(3; 4) = \text{Com}(0; 6)$, and use the binding argument to extract $\log_g h$.

Fiat-Shamir

Replace the verifier's challenge by

$$c = H(x \parallel a \parallel \text{ctx}).$$

The proof is (a, z) , or (c, z) with a recomputed.

- **Non-interactive:** a proof is a string.
- **Publicly verifiable:** anyone can check it.
- **A signature:** put the message in ctx .

ML-DSA (Lecture 18) is exactly this, for a Sigma protocol over module lattices.

Definition

H is modelled as a uniformly random function, reachable only by queries. A security proof may see every query and *choose* the answers.

- **Extraction:** find the query that hashed a , rerun with a different answer — rewinding on the oracle (forking lemma).
- **Simulation:** run the Sigma simulator for (a, c, z) , then **define** $H(x||a) := c$.

No concrete hash is a random oracle: the proof is about an idealisation.

Grinding. A cheating prover tries many a and keeps one whose challenge it can answer. With Q hash queries the error is about $Q/|\mathcal{C}|$:

$$|\mathcal{C}| = 2^{128}, \quad Q = 2^{80} \implies 2^{-48}.$$

Deniability. An interactive transcript convinces no third party; a Fiat–Shamir proof is transferable.

Not lost: zero knowledge.

Weak Fiat–Shamir

Hashing only a , omitting x , lets an attacker fix the transcript first and pick a statement to fit. Hash the statement, the commitment, the parameters and a domain separator.

From a computation to a polynomial

Goal: prove “*I ran this program and got that answer*”; the witness is an execution trace of N steps.

Index steps by a subgroup. $H = \{1, \omega, \dots, \omega^{N-1}\} \subseteq \mathbb{F}_p^\times$, ω a primitive N -th root of unity (Lecture 2). Step i is ω^i ; “next step” is $X \mapsto \omega X$.

Interpolate the trace. Each register column becomes a polynomial $T(X)$ of degree $< N$ — an inverse NTT.

Write the rules as a constraint. One polynomial $C(T(X), T(\omega X), \dots)$ with

$$\text{computation correct} \iff C(h) = 0 \text{ for every } h \in H.$$

Theorem (Lecture 2)

$Z_H(X) = \prod_{h \in H} (X - h) = X^N - 1$, and

$$C|_H \equiv 0 \iff Z_H \mid C \iff C(X) = Q(X) Z_H(X).$$

On a subgroup Z_H has two terms: the verifier evaluates it in $O(\log N)$.

Honest prover: Q exists. Dishonest prover: no polynomial Q satisfies the identity.

The protocol

Prover	Verifier
interpolates T , forms C	—
divides: $Q = C/Z_H$	—
commits to T and Q	—
—	samples $r \leftarrow \mathbb{F} \setminus H$
opens $T(r), T(\omega r), Q(r)$	recomputes $C(r)$
—	checks $C(r) = Q(r)(r^N - 1)$

A claim about N steps becomes **one field identity, checked once**.

The prover must be bound to Q *before* seeing r — otherwise it reports $Q(r) := C(r)/Z_H(r)$. That is the job of a polynomial commitment.

Theorem

Let $d = \max(\deg C, \deg Q + N)$ and r uniform on $S \subseteq \mathbb{F}$. If $Z_H \nmid C$,

$$\Pr_r [C(r) = Q(r)Z_H(r)] \leq \frac{d}{|S|}.$$

$C - QZ_H$ is nonzero of degree $\leq d$: Schwartz–Zippel (Lecture 2).

If $\mathbb{F}$ is too small for $d/|S|$ to be negligible:

- sample r from an **extension field** $\mathbb{F}_{p^k}$;
- **repeat** at independent points.

Example — arithmetise four steps

$\mathbb{F}_{17}$, $\omega = 13$ of order 4, $H = \{1, 13, 16, 4\}$, $Z_H = X^4 - 1$. One register, one rule: *it always holds a bit*. Claimed trace $(1, 0, 1, 0)$.

- a. Interpolate the trace over H to get T .
- b. Write the rule as $C = T^2 - T$ and expand.
- c. Divide by Z_H . What is Q ?

Example — the verifier's single check

Same setting.

- a. Check $C(r) = Q(r)Z_H(r)$ at $r = 3$, $r = 5$ and $r = 16$. Which r is a bad choice, and why?
- b. The prover now claims $(1, 0, 1, 2)$. Form $\tilde{C} = \tilde{T}^2 - \tilde{T}$, evaluate it on H , and show no Q exists.
- c. What is the largest degree of $\tilde{C} - QZ_H$, and the resulting soundness error over $\mathbb{F}_{17}$?

Example — how big must the field be?

$N = 2^{20}$ steps, degree-3 constraints: $d = 3 \cdot 2^{20}$.

- a. Compute $d/|\mathbb{F}|$ as a power of two for ML-DSA's prime $q = 8\,380\,417$.
- b. The same for its degree-3 and degree-5 extensions.
- c. The same for a 64-bit and a 128-bit prime.
- d. Which are usable?

Polynomial commitment scheme

$\text{Setup}(d)$; $\text{Commit}(f) \rightarrow \text{cm}$; $\text{Open}(f, r) \rightarrow (y, \pi)$; $\text{Verify}(\text{cm}, r, y, \pi)$.

Binding: no efficient prover opens cm to two values at one point.

Succinct: cm and π much smaller than f .

Arithmetisation is information-theoretic. The commitment is the cryptographic part.

Every assumption of a modern proof system lives in the polynomial commitment.

A trusted party samples a secret τ , publishes

$$\text{srs} = (g_1, g_1^\tau, g_1^{\tau^2}, \dots, g_1^{\tau^d}; g_2, g_2^\tau),$$

and **destroys** τ .

Commit

For $f(X) = \sum_{i=0}^d a_i X^i$:

$$\text{cm} = g_1^{f(\tau)} = \prod_{i=0}^d (g_1^{\tau^i})^{a_i}.$$

Computed without knowing τ . One group element, whatever d is.

To claim $f(r) = y$: r is a root of $f(X) - y$, so the division is exact.

$$q(X) = \frac{f(X) - y}{X - r}, \quad \pi = g_1^{q(\tau)}.$$

Verify (two pairings, Lecture 9)

$$e(\text{cm} \cdot g_1^{-y}, g_2) = e(\pi, g_2^\tau \cdot g_2^{-r})$$

Both sides are $e(g_1, g_2)^{f(\tau) - y} = e(g_1, g_2)^{q(\tau)(\tau - r)}$.

A wrong y' leaves a nonzero remainder: q is not a polynomial, and the srs cannot express it.

Example — the KZG quotient

$\mathbb{F}_{17}$, $f(X) = 3 + 2X + 5X^2 + X^3$, $r = 2$.

- a. Compute $y = f(2)$.
- b. Divide $f(X) - y$ by $X - 2$. What are $q(X)$ and the remainder?
- c. Claim the false value $y' = 5$. What is the remainder, and why does that stop the prover?
- d. Show by bilinearity that both sides of the verification equation are $e(g_1, g_2)$ to the same exponent.

The trusted setup

Whoever knows τ computes $(f(\tau) - y')/(\tau - r)$ as a field element and opens any commitment to any value. **It verifies.**

Mitigation: a multi-party ceremony with $\tau = \tau_1\tau_2 \cdots \tau_n$; safe if *one* participant is honest.

Not post-quantum: Shor recovers τ from the public g_1^τ . Soundness fails; zero knowledge survives.

Definition

Domain $D \subseteq \mathbb{F}_p$, $|D| = M$, degree bound $k < M$:

$$\text{RS}[D, k] = \{ (a(x))_{x \in D} : \deg a < k \} \subseteq \mathbb{F}_p^M.$$

A codeword is a polynomial in evaluation form. **Rate** k/M , **blow-up** M/k .

Minimum distance

Distinct codewords differ in at least $M - k + 1$ positions.

$a - b$ is nonzero of degree $< k$, so it has at most $k - 1$ roots.

Example — Reed–Solomon over $\mathbb{F}_{17}$

D the subgroup of $\mathbb{F}_{17}^\times$ of order 8, $k = 3$.

- a. List D . Why is it the set of nonzero squares?
- b. Give the rate, blow-up and minimum distance of $\text{RS}[D, 3]$.
- c. Encode $a(X) = 1 + 2X + 3X^2$.
- d. Build a second codeword agreeing with a at exactly two points of D ; check its distance against (b).

Commit. Evaluate f on a subgroup D , $|D| = d/\rho$ (rate $\rho = 1/4$ or $1/8$), and build a Merkle tree. The commitment is the root.

A root commits to *any* vector. FRI proves it is close to a codeword of $\text{RS}[D, d]$.

Fold. Split $f(X) = f_e(X^2) + Xf_o(X^2)$. The verifier sends random α ; the prover commits to

$$f'(Y) = f_e(Y) + \alpha f_o(Y), \quad \deg f' < d/2,$$

on $D' = \{x^2 : x \in D\}$, half the size.

Repeated folding needs $|D|$ a power of two dividing $p - 1$.

For $x \in D$, also $-x \in D$ and $(-x)^2 = x^2$:

$$f_e(x^2) = \frac{f(x) + f(-x)}{2}, \quad f_o(x^2) = \frac{f(x) - f(-x)}{2x}.$$

So $f'(x^2)$ follows from **two** evaluations of f .

- ❶ **Commit phase.** $\log_2 d$ folds; the last polynomial is a constant, sent in the clear.
- ❷ **Query phase.** s times: pick random $x \in D$, open $f_i(\pm x_i)$ and $f_{i+1}(x_i^2)$ with Merkle paths at every layer, check each fold.

Assumption: a collision-resistant hash. Nothing else.

Example — one FRI fold

$\mathbb{F}_{17}$, $D = \langle 9 \rangle$ of order 8, $f(X) = 3 + 2X + 5X^2 + X^3$ (rate $1/2$).

- a. List D and its four pairs $\pm x$. Why does $9^4 = -1$ guarantee them?
- b. Split f into f_e , f_o and fold with $\alpha = 4$.
- c. Check the fold identity at every $x \in D$ against $f'(x^2)$.
- d. Fold again with $\beta = 5$. What is the constant, and on what domain?

Query bound

If a word is at distance $\geq \delta M$ from the code, s random queries all miss with probability at most $(1 - \delta)^s$.

$M = 1024$, $k = 256$: distance 769, and 20 queries miss with probability $\left(\frac{255}{1024}\right)^{20} \approx 2^{-40.1}$.

FRI at $\rho = 1/4$, conservatively $\delta = 1/4$:

$$s = 80 \Rightarrow 2^{-33.2}, \quad s = 241 \Rightarrow 2^{-100.0}, \quad s = 309 \Rightarrow 2^{-128.2}.$$

Hundreds of queries, each with Merkle paths: **STARK proofs are kilobytes.**

SNARK	Succinct Non-interactive ARgument of Knowledge
STARK	Scalable Transparent ARgument of Knowledge

Transparent: no trusted setup. Zero knowledge is in *neither* name: it is added by hiding commitments.

Both arithmetise identically. Only the polynomial commitment differs: KZG gives a SNARK, FRI gives a STARK.

KZG has a hidden point τ ; FRI has none, so it must sample. *Transparency is paid for in queries.*

In production: prove with a STARK, then wrap its verifier in a SNARK — no longer post-quantum.

The comparison

	SNARK (KZG)	STARK (FRI)
Setup	trusted	transparent
Assumption	DLog in a pairing group, ROM	collision-resistant hash, ROM
Proof size	≈ 200 bytes, constant	tens–hundreds of KB
Verifier	a few pairings, $O(1)$	$O(\log^2 n)$ hashes
Prover	$O(n \log n)$, group operations	$O(n \log n)$, NTTs and hashes
Field	curve scalar field, $\approx 2^{254}$	two-adic prime, 2^{31} or 2^{64}
Post-quantum	no	yes

SNARK: tiny proofs, trusted setup, broken by Shor.

STARK: transparent and post-quantum, proofs $\sim 1000\times$ larger.

- 1 A proof system is complete, sound and zero knowledge; zero knowledge means a simulator, knowledge means an extractor.
- 2 Schnorr: $a = g^k$, $z = k + cx$, check $g^z = ah^c$. Extract $x = (z - z')(c - c')^{-1}$; simulate with $a = g^z h^{-c}$.
- 3 Pedersen $g^m h^r$: perfectly hiding, DLog-binding, additively homomorphic. Fiat–Shamir: $c = H(x \| a \| \text{ctx})$.
- 4 Arithmetisation: correct $\iff C = Q Z_H$, checked at one random r with error $d/|S|$.
- 5 The polynomial commitment decides the system: KZG (pairings, trusted setup) gives a SNARK; Merkle + FRI (hashes, Reed–Solomon) gives a post-quantum STARK.